

⇒ what are subarrays?

- continuous part ✓✓
- separate part
- beautiful part

$$\text{arr}[N] \rightarrow \text{subarrays} \Rightarrow \frac{N(N+1)}{2}$$

ex ⇒ 3 6 5 7 1 2 10

			subarrays
5	1	2	(X)
3	2	10	(X)
5	10		(X)

} subsequences

Subsequence :- any sequence that can be generated by deleting 0 or more element from array.

if order has to be same as array

arr ⇒ 3 6 5 7 1 2 10 ⇒ [7 2 10]

X	X	X	✓	X	✓	✓
X	✓	X	X	✓	X	X
X	X	X	X	X	X	X

⇒ [6 1]

⇒ [] ^{empty} subsequence

✓ ✓ ✓ ✓ ✓ ✓ ✓ → $\begin{bmatrix} 3 & 6 & 5 & 7 & 1 \\ 2 & 10 & & & \end{bmatrix}$
 ✓

Ques 2

arr → [1 2 3 4 5]

a) 1 2 3 4 5 ✓✓

b) 4 ✓

c) 2 3 5 ✓

d) 5 4 3 (X X) not a subsequence.

ex → arr = [-1 4 3 9]

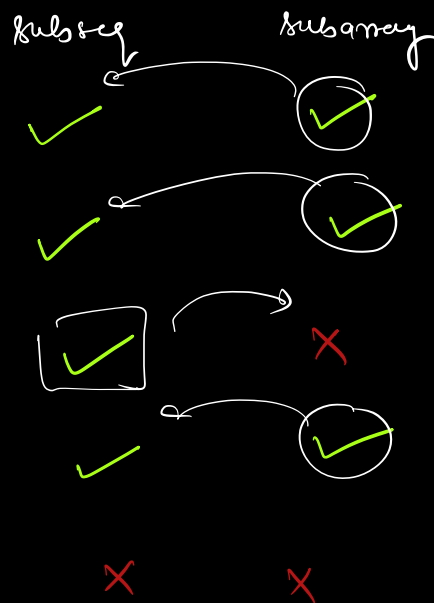
[-1 4]

[4 3 9]

[-1 3 9]

[3]

[9 3]



* all subarrays are subsequences but
vice versa is not true.

∴ A ⇒	4	-1	2	→	-1	2	4
				sort			
[] ✓	man	0			[] ✓	man	0
[4] ✓	4				[4] ✓	4	
[-1] ✓	-1				[2] ✓	2	
[2] ✓	2				[-1] ✓	-1	
[4 -1] ✗	4				✗ [-1 4]	4	
[4 2] ✗	4				✓ [-1 2]	2	
[-1 2] ✓	2				✗ [2 4]	4	
[4 -1 2] ✗	4				[-1 2 4] ✗	4	

[Subsequences will not remain same
as prior if array becomes sorted.]

∴ Subsets :-

⇒ same as subsequence but
order doesn't matter.

⇒ only unique values.

$$eq \Rightarrow arr \Rightarrow 4 \quad -1 \quad 2$$

$$\{ \} \quad \{4\} \quad \{-1\} \quad \{2\} \quad \{4, -1\} \quad \{4, 2\}$$

$$or, \{-1, 4\} \quad or, \{2, 4\}$$

$$\begin{aligned} &\{-1, 2\} \quad \{4, -1, 2\} \\ &\{2, -1\} \quad or, \{-1, 2, 4\} \\ &\quad \quad or, \{4, 2, -1\} \\ &\quad \quad \quad \vdots \\ &\quad \quad \quad \} \end{aligned}$$

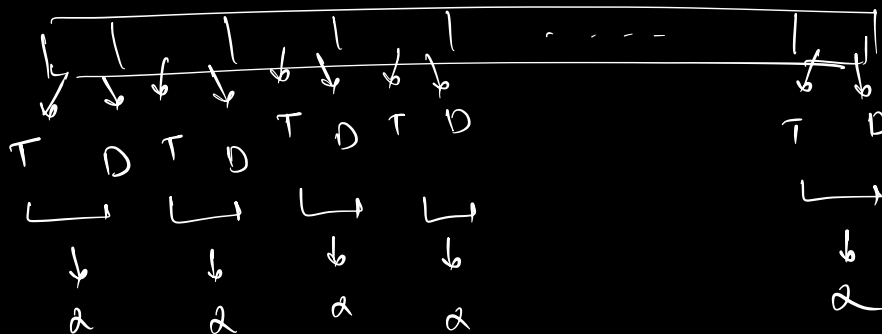
4	-1	2	$\xrightarrow{\text{sort}}$	-1	2	4
{ }			$\xleftrightarrow{\quad}$	{ }		
{4}			$\xleftrightarrow{\quad}$	{4}		
{-1}				{-1}		
{2}				{2}		
{4, -1}				{4, -1}		
{2, -1}				{2, -1}		
{4, 2}				{4, 2}		
{-1, 2, 4}				{-1, 2, 4}		

[Sorting an array will not change
all subsets of that array.]

∴ If an array doesn't contain any duplicates

no. of subsets = no. of subsequences.

⇒ Possible no. of subsequences :-



$$\left[2 \times 2 \times 2 \times 2 \times \dots \times 2 \right] \text{ N times}$$

→ 2^N

total no. of subsequences of $arr[N] \Rightarrow 2^N$.

if array has only distinct element $\Rightarrow arr[N] \Rightarrow$ no. of subsets $\Rightarrow 2^N$.

Q1, Given an array of size N (distinct elements)
check if there exists a subset with sum k .

$A \Rightarrow$

	0	1	2	3	4	5	6
	3	-1	0	6	2	-3	5

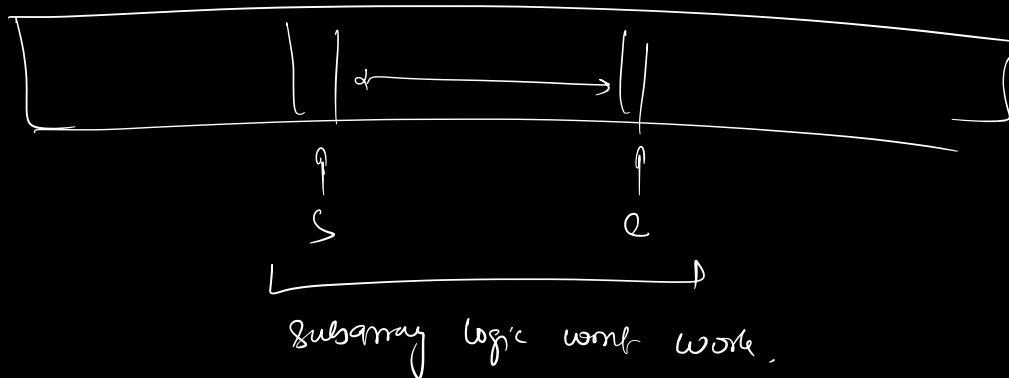
$k = 10$.

$\{5, 3, 2\} \Rightarrow 10$
 $\{6, 5, -1\} \Rightarrow 10$
 $\{6, -1, 2, 3\} \Rightarrow 10$

true

$k = 20$.

o/p $\Rightarrow$ false.



$N = 5 \downarrow$

2^5 subsets $\Rightarrow 0$ to 31 $[11111]$

eg $\Rightarrow N = 4 \downarrow$

2^4 subsets $\Rightarrow 0$ to 15 $[1111]$

$N = 3 \downarrow$

2^3 subsets $\Rightarrow 0$ to 7 $[111]$

arr $\Rightarrow$ ^{0 1 2} -2 6 4 $\Rightarrow 2^3 \Rightarrow 8$ subsets

	2	1	0	
0	0	0	0	{ }
1	0	0	1	{ -2 }
2	0	1	0	{ 6 }
3	0	1	1	{ 6, -2 }
4	1	0	0	{ 4 }
5	1	0	1	{ -2, 4 }
6	1	1	0	{ 4, 6 }
7	1	1	1	{ -2, 6, 4 }

Bit Masking

$\Rightarrow$ Check for all subset, if sum of subset $= k$, return true

arr[N] $\rightarrow 2^N \Rightarrow (0 \text{ to } 2^N - 1)$

$\rightarrow$ for ($i=0; i < 2^N; i++$) {
sum of no. for subset
 // for i^{th} subset check if sum $= k$.
 $\text{sum} = 0$
 for ($j=0; j < N; j++$) {
 if (checkBit(i, j)) {
 $\text{sum} = \text{sum} + \text{arr}[j]$
 }
 }
 }

```

    if (sum == k)
        return true
    }
    return false

```

```

for (i = 0; i < 2^N; i++) {
    sum = 0
    for (j = 0; j < N; j++) {
        if (checkBit(i, j)) {
            sum = sum + arr[j]
        }
    }
}

```

```

if (sum == k)
    return true
}

```

return false

$i = 110$
 $= 210$

arr = 2 6 4 → 0 to 7

i = 6

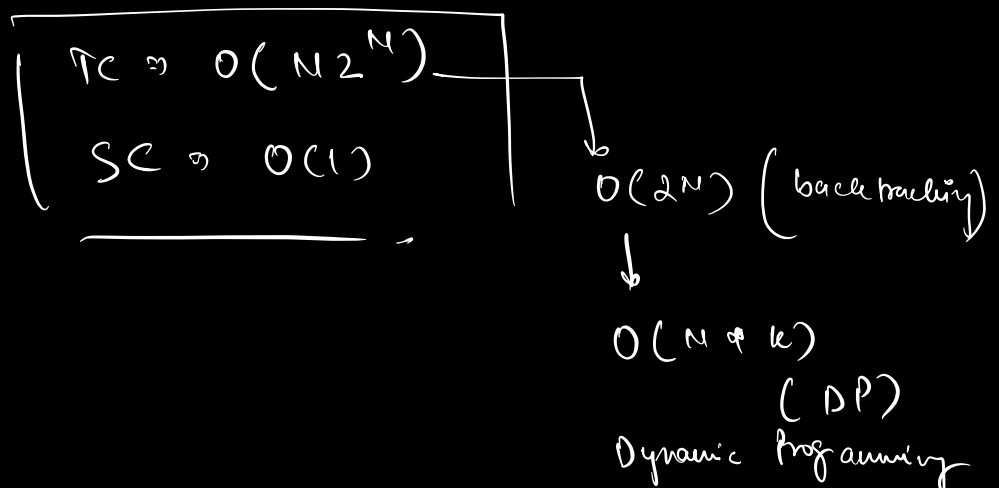
j = 0 → CB(6, 0) → X

j = 1 → CB(6, 1) → 6

j = 2 → CB(6, 2) → 4

sum = 0 ≠ 6 ≠ 4 ≠ 2

k = 10 → true



Q2. Find the sum of all subset sums, for a given array containing distinct elements.

IP $\Rightarrow \{-2, 6, 4\} \Rightarrow 8$ $2^{3-1} \Rightarrow 2^2 = \underline{4}$

$\{ \} \rightarrow 0$ Sum of all subset sum = 32

$\{-2\} \rightarrow -2$

$\{6\} \rightarrow 6$

$\{4\} \rightarrow 4$

$\{-2, 6\} \rightarrow 4$

$\{6, 4\} \rightarrow 10$

$\{-2, 4\} \rightarrow 2$

$\{-2, 6, 4\} \Rightarrow 8$

$\{ \} \rightarrow 0$

Optimized

$\{-2\} \rightarrow -2$

$\{6\} \rightarrow 6$

$\{4\} \rightarrow 4$

$\{-2, 6\} \rightarrow 4$

$\{6, 4\} \rightarrow 10$

$\{-2, 4\} \rightarrow 2$

$\{-2, 6, 4\} \Rightarrow 8$

Brute force

using solⁿ of previous problem

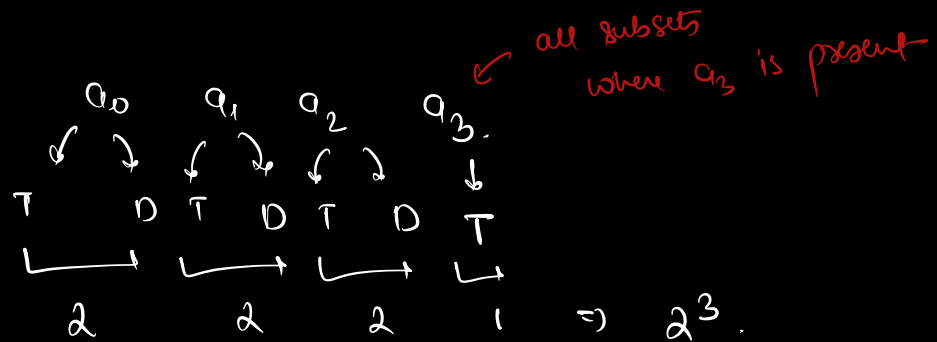
TC $\Rightarrow O(N 2^N)$
 SC $\Rightarrow O(1)$

Sum $\Rightarrow -2 \times 4 + 6 \times 4 + 4 \times 4$

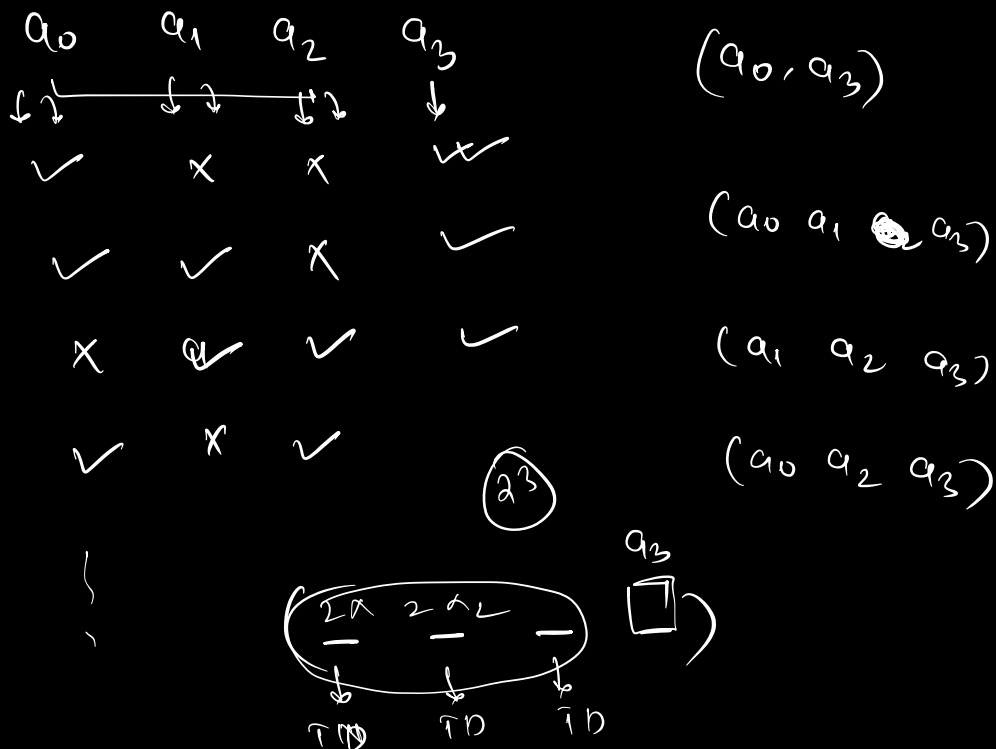
Total Sum = Sum of all subset sums

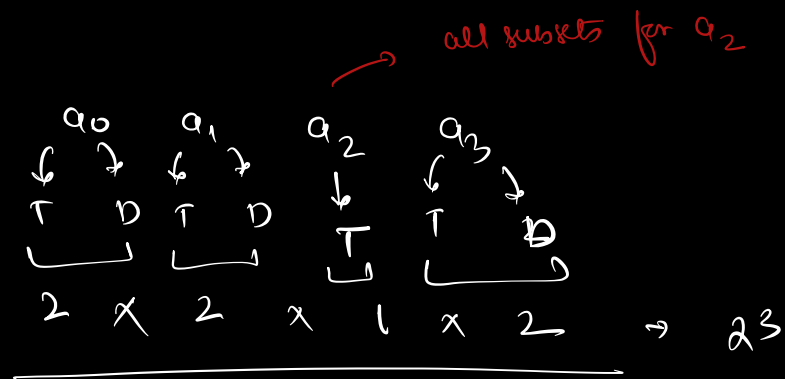
⇒ to find the contribution of a no. in total sum,
we need to find in how many subsets that element
is present.

Ans ⇒



a_3 is present in 2^3 subsets.





$$\begin{cases}
 a_0 \rightarrow 2^3 \\
 a_1 \rightarrow 2^3 \\
 a_2 \rightarrow 2^3 \\
 a_3 \rightarrow 2^3
 \end{cases}$$

any element i in an array of length N is present in 2^{N-1} subsets.

pseudo

sum = 0

for ($i=0; i < N; i++$) {

sum = sum + ($arr[i] \times 2^{N-1}$)

}

return sum;

$\downarrow$
 $1 \leq N-1 \Rightarrow O(1)$

TC $\Rightarrow O(N)$
 SC $\Rightarrow O(1)$

Zeta

Q3. Given an arr of size N containing distinct elements
Calculate $(\text{sum of all subset sum}) / 2^N$

$$\text{arr}[N] \rightarrow a_0 \quad a_1 \quad a_2 \quad \dots \quad a_{N-1}$$

$$\text{sum of all subsets} \Rightarrow a_0 \times 2^{N-1} + a_1 \times 2^{N-1} + a_2 \times 2^{N-1} \dots + a_{N-1} \times 2^{N-1}$$

$$\text{sum} \Rightarrow 2^{N-1} (a_0 + a_1 + a_2 + \dots + a_{N-1})$$

$$\left(\text{sum of all subset sums} \right) / 2^N \Rightarrow \frac{2^{N-1} (a_0 + a_1 + a_2 + \dots + a_{N-1})}{2^N}$$

$$\Rightarrow \frac{\cancel{2^{N-1}} (a_0 + a_1 + a_2 + \dots + a_{N-1})}{2 \times \cancel{2^{N-1}}}$$

$$O/P \Rightarrow \frac{(\text{sum of all arr. elements})}{2}$$

$$\begin{aligned} TC &\Rightarrow O(N) \\ SC &\Rightarrow O(1) \end{aligned}$$

facebook

Q4. Given an array, find sum of max of all subsequences

ex $\Rightarrow$ $A \Rightarrow 3 \ 1 \ -4$

$[]$ ^{max}
0

$[3]$ 3

$[1]$ 1

$[-4]$ -4

$[3 \ 1]$ 3

$[3 \ -4]$ 3

$[1 \ -4]$ 1

$[3 \ 1 \ -4]$

sum $\Rightarrow$ 10.

$\longleftarrow$

$3 \times 4 + 1 \times 2 + (-4) \times 1$

Brute force

for all subsequences find max & sum

$$\boxed{\begin{array}{l} TC \Rightarrow O(N 2^N) \\ SC \Rightarrow O(1) \end{array}}$$

Optimised

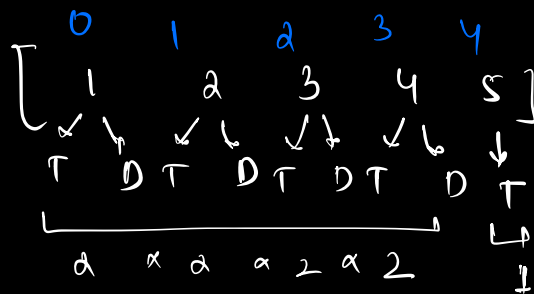
Contribution technique

⇒ [3 2 1 4 5]

↙
sort [1 2 3 4 5]

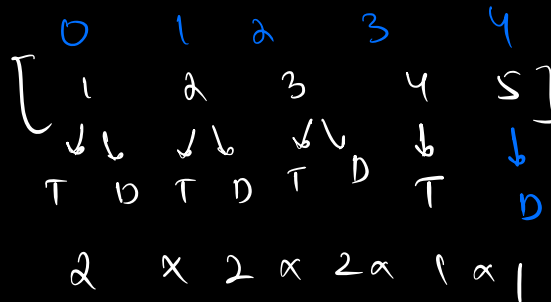
no. of subsequences
where 5 is max^m

↓
 2^4



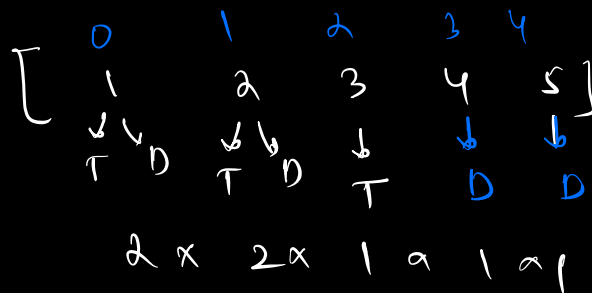
no. of subseq.
where 4 is max^m

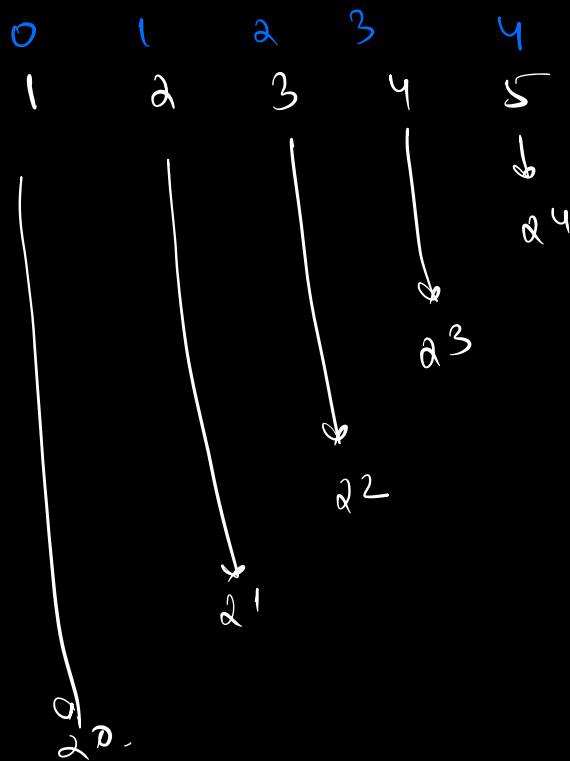
↓
 2^3



no. of subseq
where 3 is max^m

2^2





pseudo

→ sort the array

→ sum of all contribution

→ contribution for element $arr[i] \rightarrow 2^i$ times

sort()

sum = 0

for ($i=0; i < N; i++$) {

sum = sum + $arr[i] \times 2^i$

}

return sum

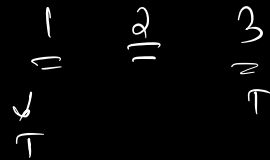
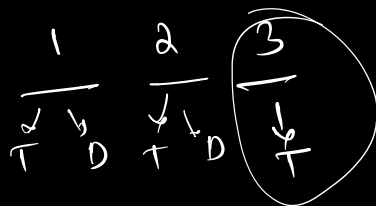
TC $\approx O(N \log N)$

SC $\approx O(1)$

$1 \leq i$

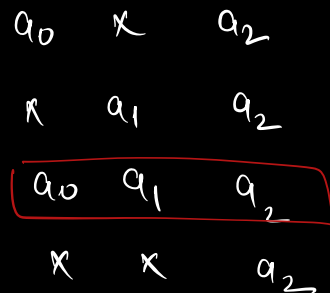
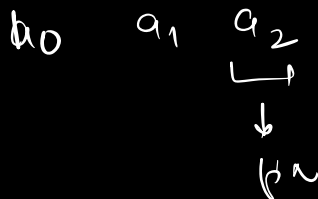
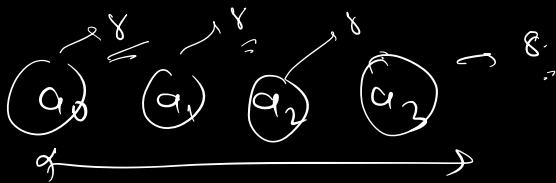
- i) Revise $\rightarrow$ intermediate $[24]$
 $\downarrow$
8
- ii) 80% PSP
- iii) Revise $\rightarrow$ inter notes for
each topic prior to starting
that topic in advance

Doubts



1 2 3

1 2 3



a_0 a_1 a_2
 └─┘
 for

x a_1 a_2

a_0 a_1 x

a_0 a_1 a_2

x a_1 x

$\downarrow$ $\downarrow$ $\downarrow$
 a_0 a_1 a_2

$2^N \Rightarrow$

a_0 a_1 a_2
 $\downarrow \downarrow$ $\downarrow \downarrow$ $\downarrow \downarrow$
 T D T D T D

$\textcircled{8}$
1

a_0 a_1 a_2

a_0 x a_2
 x a_1 a_2
 a_0 a_1 a_2
 x x a_2

$Pow(2, N-1) \rightarrow \log N$
 $1 \leq (N-1) \Rightarrow O(1)$

a_0 a_1 a_2